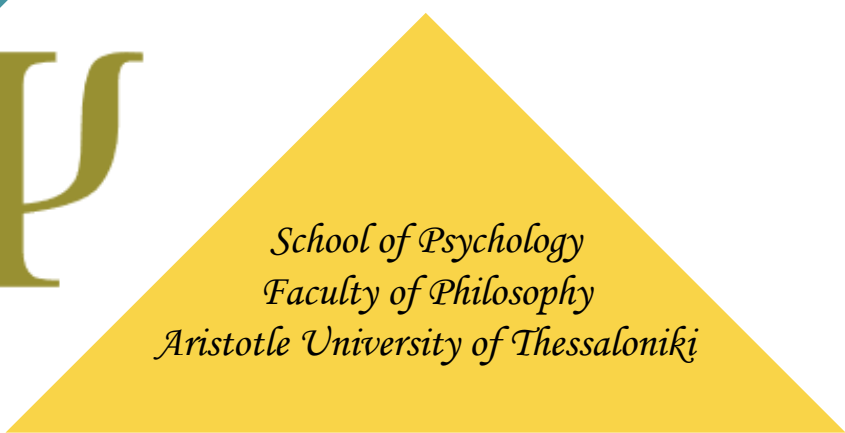




Linear Regression Models

Simple models

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What do we mean by a statistical model?

- A simplification or approximation of reality. (Burnham, Anderson, 2002)
- Statistical models summarize patterns of the data available for analysis. (Steyerberg, 2009)
- A powerful tool for developing and testing theories by way of causal explanation, prediction, and description. (Shmueli, 2010)

Basic Properties

- They should be **valid**: provide explanations or predictions with acceptable accuracy
 - They should be **practically useful**: allow conclusions such as “how large is the expected change in outcome if one of the explanatory variables changes by one unit”
 - They should be **robust**.
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To Explain or to Predict?

Modeling for explanation

Describe and quantify the association between the outcome variable Y and a set of explanatory variables X 's.

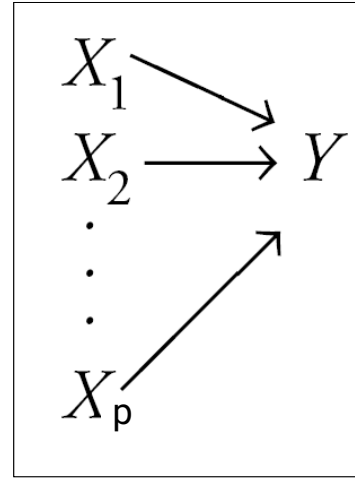
- Identification of 'important' explanatory variables
- Understanding the effects of explanatory variables
- Adjustment for variables uncontrollable by experimental design

Modeling for prediction

When we want to predict an outcome variable Y based on the information contained in a set of predictor variables X 's.

Linear Regression model

The effect of one or more (**continuous or categorical**) independent variables X_p on the values of a **continuous** dependent variable Y .



Example:

We would like to examine whether several variables (e.g., **height, headc, gender, parity, education**) have an effect on **weight** (in g) of infants at 1-month age.

Basic assumptions

Linearity: linear combination of variables

- (Relaxation: splines, fractional polynomials, etc.)

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \cdots + \hat{\beta}_p x_{pi}$$

Additivity: sum of main effects

- (Relaxation: include interactions etc.)

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_{1,2}(\mathbf{x}_{1i} \cdot \mathbf{x}_{2i}) + \cdots + \hat{\beta}_p x_{pi}$$

Data

The data of 550 infants at 1 month age were collected (**BirthWeight**). The following variables were recorded:

- Body weight of the infant in g (**weight**)
- Body height of the infant in cm (**height**),
- Head circumference in cm (**headc**),
- Gender of the infant (**gender**: Female, Male)
- Birth order in their family (**parity**: Singleton, One sibling, 2 or more siblings)
- Education of the mother (**education**: year10, year12, tertiary)

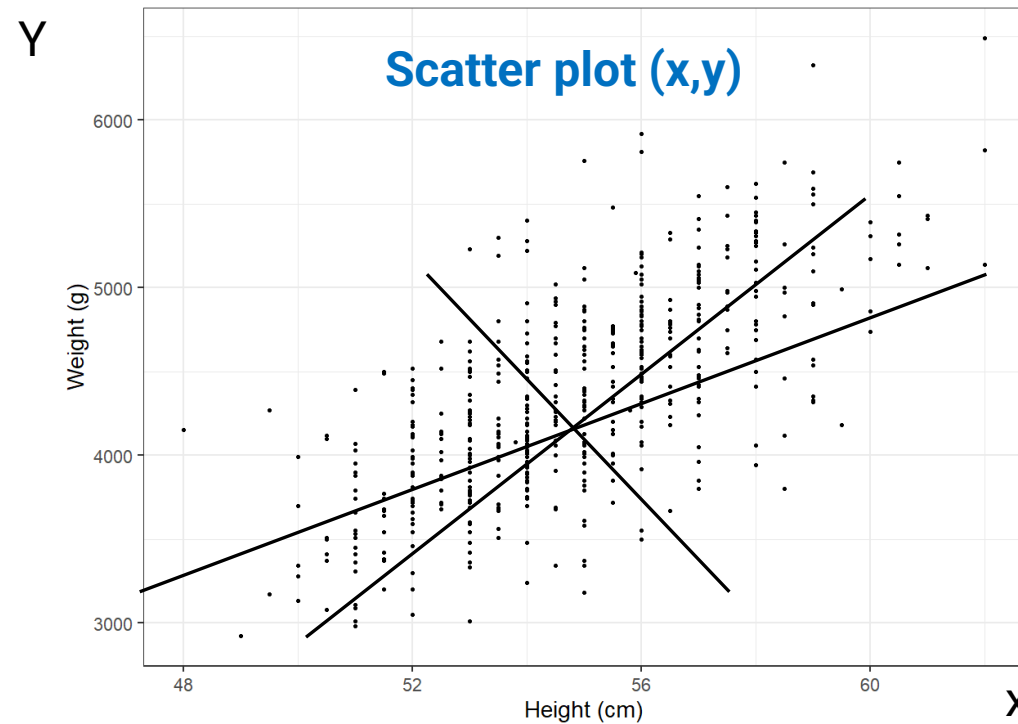
Simple Linear regression

Dependent variable $\rightarrow y_i = \beta_o + \beta_1 * x_i + \varepsilon_i \quad i=1,2,\dots,n$

Independent variable $\rightarrow x_i$

Systematic component ($\hat{y}_i$) $\rightarrow \beta_o + \beta_1 * x_i$

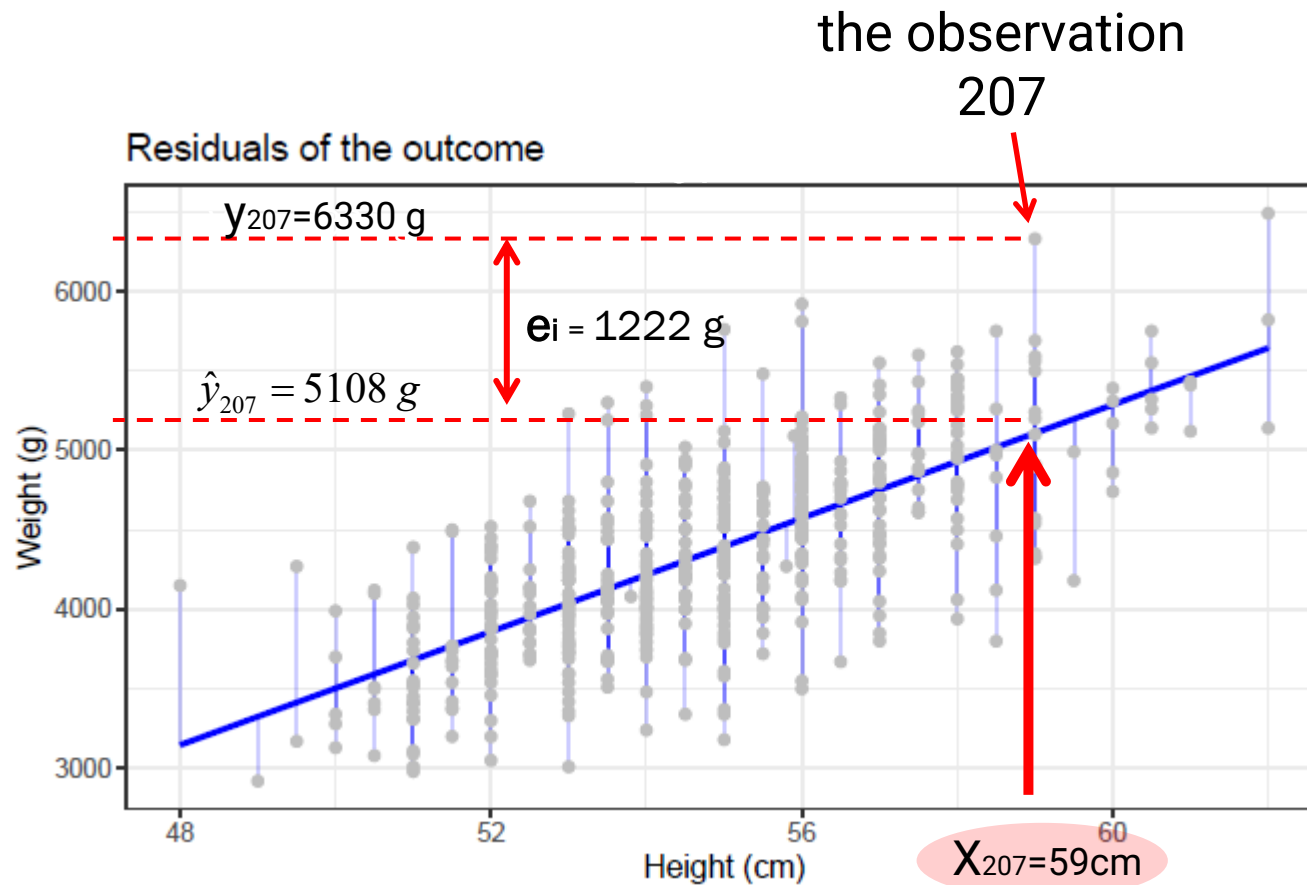
Random error $\rightarrow \varepsilon_i$



X: height
(independent or explanatory variable)

Y: weight
(response or dependent variable)

Line of best fit (direct regression)



Residuals (error)

$$\hat{e}_i = y_i - \hat{y}_i$$

$$\sum_{i=1}^n \hat{e}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 * X_1)^2$$

least squares estimates

$$\hat{\beta}_0, \hat{\beta}_1$$

Best fitted line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 * x_i$$

Continuous explanatory variable

Question:

What is the association between **weight** and **height** ?



Hypothesis Testing

$$\hat{weight} = \hat{\beta}_0 + \hat{\beta}_1 * height$$

- **H₀: $\beta_1=0$** (no association)
- **H₁: $\beta_1 \neq 0$** (there is association)

Results and interpretation

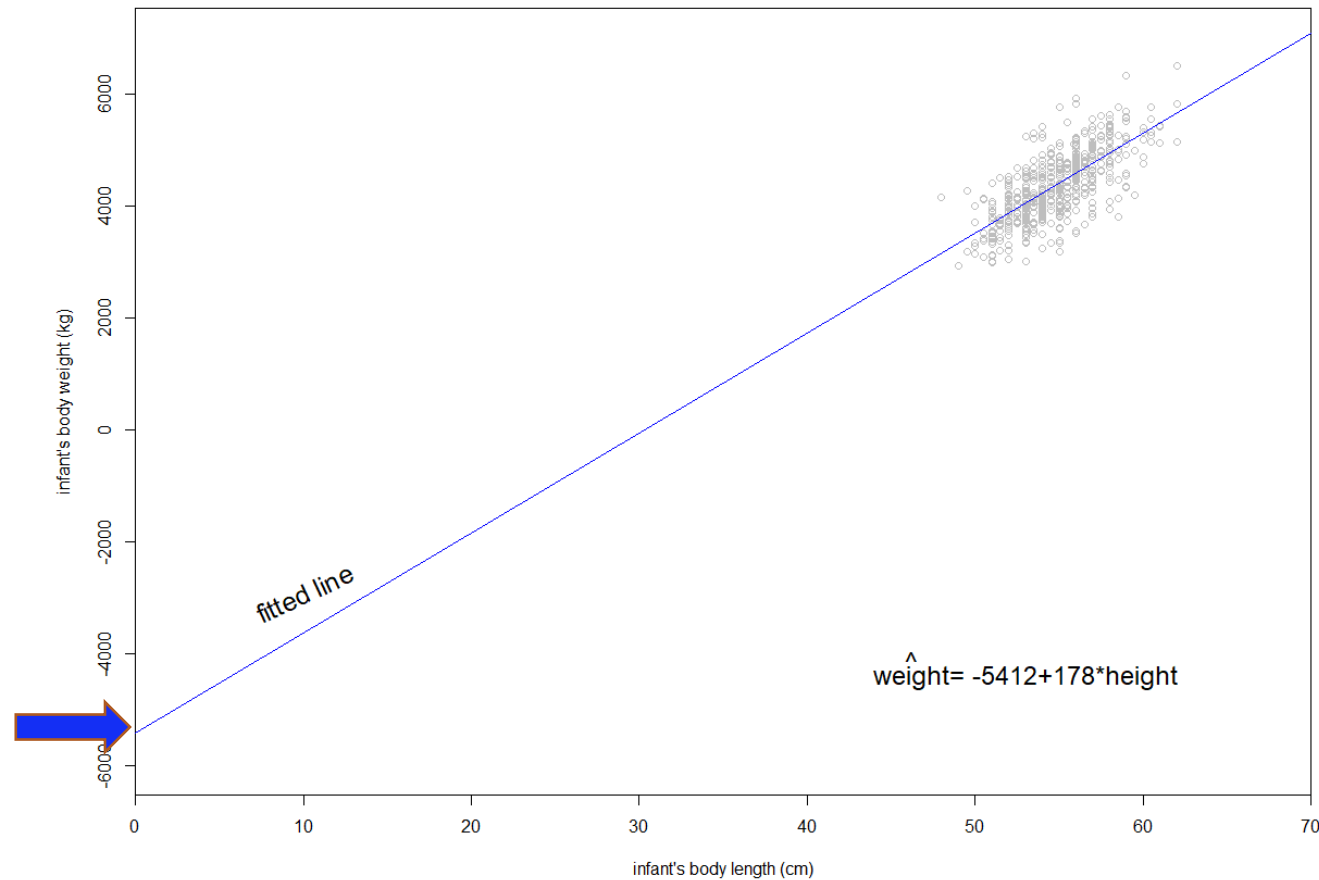
$$\hat{weight} = -5412 + 178 * height$$

On average, there's an expected increase of **178** g of weight for **every 1 cm increase** in height (95%CI: 164 to 193, P<0.001)

The intercept

$$\hat{weight} = -5412 + 178 * height$$

Plot the fitted line **crossing the y-axis** (weight):



The fitted line crosses the y-axis roughly at -5400.

This value is the estimate of the intercept β_0 . **Not physical interpretation.**

The slope



The **slope** β_1 from two points of the fitted line is:

$$\hat{\beta}_1 = \frac{\Delta y}{\Delta x} = \frac{5270 - 4560}{60 - 56} = \frac{710}{4} \approx 178 \text{ g/cm}$$

Binary explanatory variable

Question:

What is the association between **weight** and **gender** of the infant?

$$\text{genderMale} = \begin{cases} 1 & \text{if infant is Male} \\ 0 & \text{otherwise (reference)} \end{cases}$$

$$\widehat{\text{weight}} = b_o + b_1 \cdot \text{genderMale}$$

Results and interpretation

$$\text{genderMale} = \begin{cases} 1 & \text{if infant is Male} \\ 0 & \text{otherwise (reference)} \end{cases}$$

$$\widehat{\text{weight}} = 4140 + 452 \cdot \text{genderMale}$$

- For **females**:

$$\text{Weight} = 4140 + 452 \cdot 0 = 4140 \text{ g}$$

The intercept is the mean body weight (in g) for a female infant which is the **reference category**.

- For **males**:

$$\text{Weight} = 4140 + 452 \cdot 1 = 4140 + 452 = 4592 \text{ g}$$

The coefficient value 452 is **the difference** (4592 – 4140) in the **mean** weight (in g) for a male infant **relative** to a female infant.

Conclusion

The mean weight of a male infant is 4592 g which is **significantly higher about 452 g** relative to a female infant of 4141 g (95%CI: 358 to 545, $p < 0.001$)

The above analysis is equivalent to perform a **two-sample t-test!**

Decorative geometric shapes in teal, yellow, and green are located in the bottom-left corner of the slide.

Categorical explanatory variable (>2 categories)

Question:

What is the association between **weight** and **birth order** in the family (parity) of the infant?

$$parity = \begin{cases} \textit{Singleton (ref.)} \\ \textit{One sibling} \\ \textit{2 or more siblings} \end{cases}$$

Dummy variables

A categorical explanatory variable with **k-levels** or categories requires **(k-1) dummy variables** to represent it.

→ The explanatory variable, **parity**, has three categories, so we need to create **two dummy variables**.

Dummy variables

Considering the **Singleton** as the reference group:

$$\text{parityOne sibling} = \begin{cases} 1 & \text{if infant has one sibling} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{parity2 or more siblings} = \begin{cases} 1 & \text{if infant has 2 or more siblings} \\ 0 & \text{otherwise} \end{cases}$$

parity	One sibling	2 or more siblings
Singleton (ref.)	0	0
One sibling (parity1)	1	0
2 or more siblings (parity2)	0	1

We are including all the categories to the linear regression model **except one which is going to be used as the reference group** (here the Singleton category).

Results and interpretation

$$\widehat{\text{weight}} = 4259 + 130 \cdot \text{parityOne sibling} + 192 \cdot \text{parity2 or more siblings}$$

- For a singleton infant:

$$\text{Weight} = 4259 + 130 \cdot 0 + 192 \cdot 0 = 4259 \text{ g}$$

The **intercept** equals to the mean weight in g for a singleton infant **which is the reference category**.

- For an infant with **one sibling**:

$$\text{Weight} = 4259 + 130 \cdot 1 + 192 \cdot 0 = 4259 + 130 = 4389 \text{ g}$$

The coefficient for “One sibling” dummy variable is **130** and represents the difference in the mean weight in grams for an infant with **one sibling relative to a singleton infant**.

- For an infant with **2 or more siblings**:

$$\text{Weight} = 4259 + 130 \cdot 0 + 192 \cdot 1 = 4259 + 192 = 4451 \text{ g}$$

The coefficient for “2 or more siblings” dummy variable is **192** and represents the difference in the mean weight in grams for an infant with **2 or more siblings relative to a singleton infant**.